What a Robot Can Measure, It Can Undo: Separating Invertible Nuisances from Policy Failures at Deployment

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Abstract

A manipulation policy that works in the lab fails when the camera is moved, the arm starts somewhere new, or the light changes. Some of those failures are the policy's; others are only a change of viewpoint or pose that could be undone before the policy ever sees it. We ask which is which, before any rollout and without touching the policy, using the robot's own body as the instrument: its encoders, its kinematic model and its silhouette in the image. From these we compute a recoverability score, recover = ident × invert: can the robot measure the nuisance, and would undoing it restore the training observation? Frozen before its test, the score predicts which failures a training-free correction repairs on two held-out policy families (AUROC 0.81 and 0.68), where the perturbation category, the label robustness benchmarks report, is at chance (0.52 and 0.47). The same instrument then does the undoing. Servoing the arm to its training pose from proprioception repairs 47 of 47 failures on three suites nothing was tuned on; a camera pose read from the robot's silhouette, with no markers, repairs 61 of 100 camera failures across four suites ($p = 1.3 \times 10^{-6}$) and is significant on the held-out $\pi_{0.5}$ over replicated rollouts. Measuring only when the silhouette says the camera moved turns the wrapper from a wash into a net gain on that policy ($p = 0.013$). Attributing every remaining camera failure by intervention, on the development policy 1 in 100 belongs to the policy; the rest are measurement or lost view, which the score flags in advance.

1 Introduction

Vision-language-action (VLA) policies are evaluated under one condition and deployed under another. LIBERO-Plus (Fei et al., 2025) made the gap concrete: moving the camera or the robot's initial pose takes policies that solve a task nine times in ten to below three in ten. The response has been to make the policy robust, or to wrap it: restore the corrupted stream (Yan & Shen, 2026; Guarnizo Orjuela et al., 2026; Hancock et al., 2025), re-render the view (Heo et al., 2026; Gu et al., 2026; Park & Kim, 2026), or predict the camera inside the policy (Li et al., 2026).

This paper asks a prior question. A deployment shift is either a nuisance, a change to how the scene is observed that leaves the task untouched and could in principle be inverted, or it changes the task. Benchmarks label perturbations by category, and wrappers route corrections by that label. The label does not tell you what is fixable: a camera that orbits 45 degrees is recoverable and a camera that tilts 35 degrees away is not, and both are "camera". What decides it is whether the robot can measure the change and whether undoing it restores what the policy was trained on. Both can be computed by the robot, about itself, before the policy acts.

The body as the instrument. A robot carries a calibrated object everywhere it goes: itself. Encoders give its pose; a kinematic model and a renderer give where its silhouette must appear for any camera pose. We use this one instrument to measure the nuisance, predict

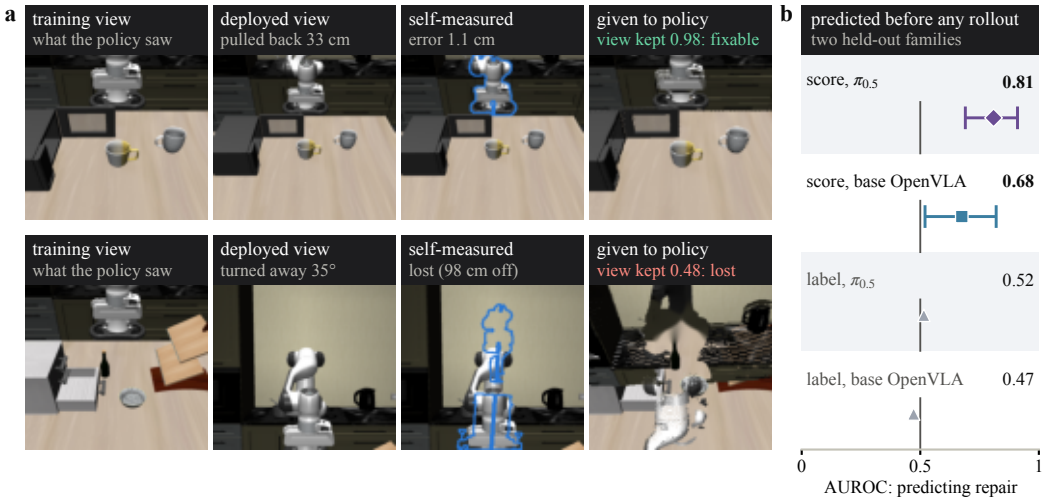


Figure 1: The wrapper changes no weights. (a) Two camera shifts. The robot measures the camera from its own body (blue: its silhouette rendered at the measured pose) and re-renders the frame into the training view for the frozen policy. How much of the view survives is known before any rollout: the pull-back can be undone, the look-away cannot. (b) The policy-free score predicts which failures are repaired on two held-out policy families, with 95% intervals; the perturbation category is at chance (Table 1).

whether undoing it will work, undo it, and verify the undo, all with a frozen policy and no ground truth.

Contributions. (1) A policy-free recoverability score, frozen before its held-out tests, that predicts which failures are repairable on two families it never saw, where the category label is at chance (Sec. 4.1). (2) Correction from self-measurement: a proprioceptive servo to the training pose, which no prior wrapper uses and which repairs 47 of 47 failures on untuned suites, and a marker-free camera pose from the robot's silhouette that drives a depth reprojection (Sec. 4.2). (3) A motion-free test for whether to measure at all (Prop. 3), which removes the main cost of self-measurement on a held-out policy and makes the wrapper a net gain there (Sec. 4.3). (4) Where deployment failures live: an interventional ladder that assigns 1 of 100 camera failures of the development policy to the policy itself, a measured calibration tolerance of about a centimetre, and the finding that a behaviour-cloned policy's opening action field diverges, which rules out goal-triangulation probes and motivates an action-field fingerprint (Sec. 4.4, 4.5).

2 Related work

Wrappers around frozen VLAs. Restoring the Right Stream (Yan & Shen, 2026) is the closest work: a training-free per-stream restoration on the same benchmark, routed by stream identity; our category-label baseline is exactly such a routing signal, and it is at chance on what can be repaired. BYOVLA (Hancock et al., 2025) edits task-irrelevant image regions, CRT (Guarnizo Orjuela et al., 2026) restores corrupted inputs, and ProbeAct (Zhang et al., 2026) repairs actions rather than observations. Viewpoint. AnyCamVLA (Heo et al., 2026) re-renders the test view for a frozen VLA with a pose from fiducial markers; VistaBot (Gu et al., 2026) and GS-VLA (Park & Kim, 2026) learn the synthesis; CamVLA (Li et al., 2026) regresses the camera inside a retrained policy. Reprojection is not our contribution; the source of the pose (the robot's own kinematics and silhouette, no marker, no learned head) and saying in advance when it can work are. Robot as calibration target. Render-and-compare robot pose estimation (Lee et al., 2020; Labbé et al., 2021; Lu et al., 2023; Bilić et al., 2024) and mask-based hand-eye calibration (Chen et al., 2023; Hong et al., 2024) are established, and DROID (Khazatsky et al., 2024) has been calibrated offline from robot masks (Xie et al., 2025; Irshad et al., 2025); we claim no calibration algorithm, and

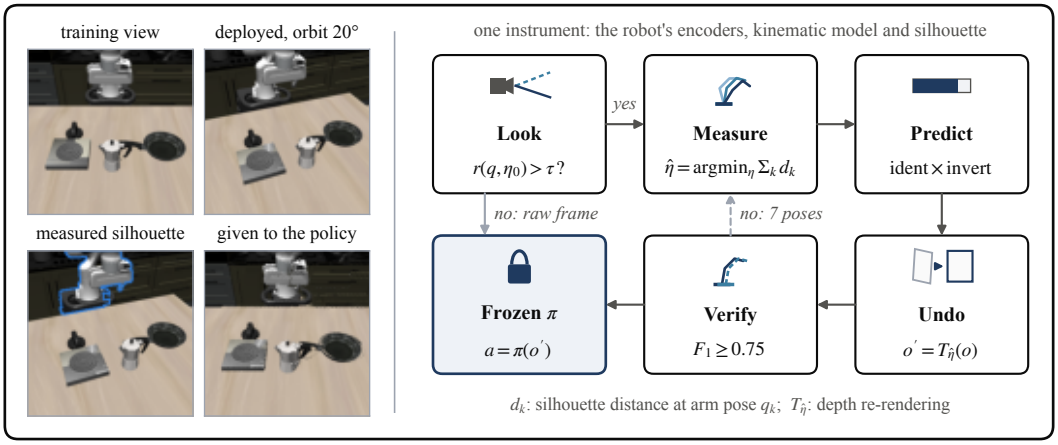


Figure 2: The wrapper around a frozen policy. Left: one task in simulation (camera orbited 20 degrees): the training view, the deployed view, the robot's silhouette rendered at the camera pose it measured from its own body (blue), and the frame re-rendered into the training camera. Right: the loop. Look was added after the main experiments (Sec. 4.3); if the camera has not moved, the arm stays still and the policy gets the raw frame.

use one to measure displacement from the training view at deployment. Predicting failure. RoboART (Majumdar et al., 2025) predicts a policy's degradation under visual shift with the policy's own embedding; we predict repairability without the policy. Runtime monitors (Agia et al., 2024; Xu et al., 2025; Gu et al., 2025; Römer et al., 2025; Huang et al., 2026; Yang et al., 2026) read statistics of the actions a policy emits; our fingerprint queries the policy at poses the robot chooses. Task-relevant shift (Farid et al., 2021), identifiability for test-time adaptation (Jhawar & Wang, 2026), selective adaptation (Jiang et al., 2026) and learned canonicalisation (Kaba et al., 2023) are the closest conceptual ancestors; error amplification in closed-loop imitation (Block et al., 2024) and small-sample policy evaluation (Vincent et al., 2024) frame our rollout statistics.

3 Measure, predict, undo, verify

Let $o = g(s;\eta)$ be the observation of state s under nuisance parameters η (camera pose, initial joint configuration, illumination, sensor noise), η_0 the training setup, and π a frozen policy. A nuisance is invertible at η if a map T_η, computable from the robot's own measurements, gives $T_\eta(g(s;\eta)) \approx g(s;\eta_0)$ on the states the task visits. Fig. 2 draws the loop.

Measure. The robot records self-observations z that need no knowledge of the scene: joint readings, the silhouette of its arm at a few end-effector poses it visits deliberately, the colour statistics of its own links, and the noise level of its own pixels. The camera pose $\hat{\eta}_c$ minimises a silhouette chamfer plus a size term over the visited poses (coarse orbit grid, cross-entropy refinement, depth ICP) and is accepted only if it beats the training pose by a fixed factor; light and noise are an affine colour fit and a noise floor on the robot's own pixels.

Predict.

$$\text{recover}(\eta) \;=\; \underbrace{\min\!\left(1,\; \frac{\sigma_{\min}(J_\eta)}{\sigma_{\min}(J_{\eta_0})}\right)}_{\text{ident}} \;\times\; \underbrace{\text{invert}(\eta)}_{\text{does undoing restore the training view?}}$$

$J_\eta = \partial z / \partial \eta$ is the Jacobian of the self-observations with respect to the nuisance, so ident asks whether the body still responds to the nuisance as it did in the training setup. invert is per channel and also self-measured: for a camera, the fraction of the training view the reprojection from $\hat{\eta}_c$ can fill (coverage); for light, an affine-invariant shading statistic of the robot's links; for noise, the share of added noise the denoiser removes on the robot's pixels; 1 for joint offsets; 0 for scene content, which has no inversion. Every constant was frozen and logged before the first episode of either held-out family.

Undo and verify. Joint offset: servo to the training pose. Camera: reproject into the training camera with $\hat{\eta}_c$ and depth, z-buffered. Light: invert the affine fit unless the shading statistic says the change is not affine. Noise: denoise at the measured level. No weights change and no policy gradient is used. After a camera correction the robot warps its observed silhouette with the same pose and depth and compares it with where its kinematics place it in the training view; if they disagree ($F_1 < 0.75$, fixed from policy-free data) it measures again with seven poses and keeps the calibration its silhouette agrees with more.

3.1 What can be proved

Four statements carry the method; proofs are in Appendix D, and each is tested by one experiment. Let $z(q, \eta)$ be the self-observation at joint configuration q, with the robot choosing $q_1, \ldots, q_K$.

Proposition 1 (measurement error from the body alone). If $J = \partial[z(q_1, \eta); \ldots; z(q_K, \eta)]/\partial\eta$ has full column rank and the self-observations carry independent noise of variance σ^2, every unbiased estimator satisfies $\mathbb{E}\|\hat{\eta} - \eta\|^2 \geq \sigma^2 \operatorname{tr}\big((J^\top J)^{-1}\big) \geq \sigma^2/\sigma_{\min}(J)^2$, and adding a pose never lowers $\sigma_{\min}(J)$.

This is why ident is a ratio of smallest singular values. Measured on 8 camera tasks: $\sigma_{\min}$ rises from 1.04 to 3.58 from one pose to seven, and the calibration error falls from 0.54 to 0.07 cm.

Proposition 2 (what re-rendering can restore). Let V be the training-view pixels visible from the deployed camera, coverage $c = |V|/|\Omega|$. Any restoration $\hat{o}$ computed from the deployed frame alone satisfies $\mathbb{E}\,\|\hat{o} - o_{\text{train}}\|_1 \geq (1 - c)\, e_{\text{prior}} + c\, e_V$, with e_{prior} the best prior fill and e_V the resampling error on seen pixels: no single-view re-rendering wrapper, ours or any other, restores what the deployed camera did not see.

Coverage is therefore an upper bound on the camera invert, not a heuristic.

Proposition 3 (a motion-free test for "the camera moved"). If the camera is at η_0 and the body model is exact, the residual $r(q) = d\big(z_{\text{obs}}(q),\, z(q, \eta_0)\big)$ is zero up to noise; a residual above the noise floor proves $\eta \neq \eta_0$. One pose does not identify η, but it decides whether identification is needed.

Proposition 4 (a diverging opening field has no sub-goal). Let the opening action near p_0 be $a(p) = a_0 + A\,(p - p_0)$. Triangulating a sub-goal from probes assumes $A = -\lambda I \prec 0$. If the symmetric part of A is positive definite, the least-squares point is a source the field flows away from, and its residual carries no information about miscalibration.

4 Experiments

Setup. LIBERO-Plus (Fei et al., 2025), four suites. Policies: OpenVLA-OFT (Kim et al., 2025) with its per-suite checkpoints (development family), and two held-out families on which no constant was chosen: $\pi_{0.5}$ (Physical Intelligence et al., 2025) and base OpenVLA (Kim et al., 2024), a much weaker policy (0.19 over 210 LIBERO-Plus tasks). All comparisons are paired, the same task and seed under the raw and the wrapped policy, with exact McNemar tests on discordant pairs. Simulator segmentation and depth stand in for an RGB-D sensor and a robot segmenter (Sec. 5). Analyses designed after the main results are marked †. Tests registered before their data existed, and their outcomes, are listed in Appendix A.

4.1 The recoverability score

The score transfers to both held-out families while the category label stays at chance (Table 1). On $\pi_{0.5}$ identifiability alone reaches 0.53; the six look-away conditions, which the label calls recoverable, score 0.14 to 0.49 and none is repaired. Over 264 camera placements and every LIBERO-Plus camera task that logged it (200 tasks, 88 raw failures, two families), the camera term is calibrated: observed repair rises from 14% to 88% with the predicted value (AUROC 0.78 [0.67, 0.88]; Fig. 3).

Table 1: Predicting, without the policy, which raw failures the correction repairs (graded sweep: 24 conditions on base tasks each policy solves unperturbed). One set of constants, frozen before either held-out family was run. The first three-task test is in Appendix C.

	units (raw failures)	repaired	AUROC, score	AUROC, category label
OpenVLA-OFT (development), ten base tasks	65	37	0.823 [0.71, 0.92]	0.518
$\pi_{0.5}$ (held out), ten base tasks	72	37	0.809 [0.69, 0.91]	0.515
base OpenVLA (held out), four base tasks	60	24	0.675 [0.52, 0.82]	0.472
pooled	197	98	0.763 [0.69, 0.83]	0.510

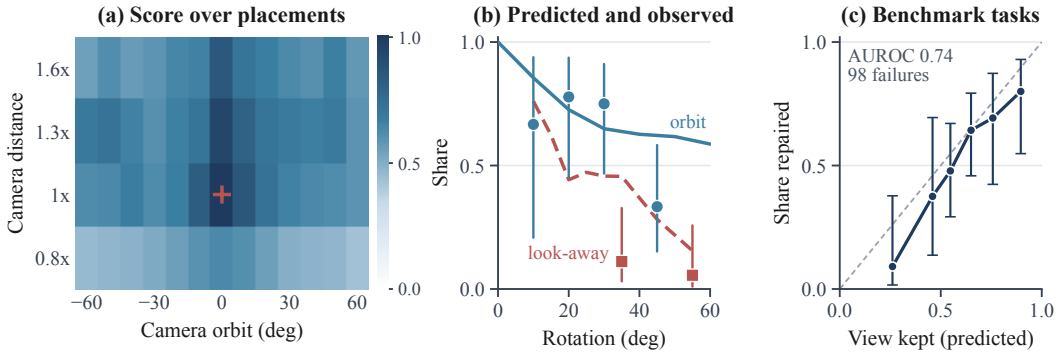


Figure 3: The score before any policy runs, and what happened when one did. (a) The score over 264 camera placements around the training camera (cross), three base tasks. (b) Along an orbit and a look-away: the score (lines) and the share of raw failures the wrapper repaired on the sweep (points, OFT and $\pi_{0.5}$ pooled, 95% Wilson intervals). (c) Every LIBERO-Plus camera task that logged the camera term (four OFT suites and $\pi_{0.5}$): share repaired against the view the robot measured it would keep; dashed: perfect calibration.

Why the weaker family scores lower†. "Repaired" means the raw policy failed and the wrapped one succeeded. On base OpenVLA, 5 of its 24 repaired units had no channel applied at all (OFT: 0 of 37; $\pi_{0.5}$: 1 of 37): the measurement's arm motion jostled a fragile policy into success. Four are background conditions, where the score correctly assigns invert = 0 and is scored as wrong for it. Excluding those units the three families agree: 0.823, 0.831 and 0.816. On a fragile policy the label, not the score, carries the noise, and Sec. 4.3 removes its source.

4.2 Undoing what was measured

Joint offsets. The proprioceptive servo is the most reliable repair we measure: 47 of 47 failures on the three untuned suites with 1 of 13 successes broken ($p = 3 \times 10^{-13}$), and 16 of 19 with none broken on $\pi_{0.5}$. Camera. Over the four suites the wrapper repairs 61 of 100 camera failures and breaks 18 of 140 successes ($p = 1.3 \times 10^{-6}$); it is significant alone on object ($p = 5 \times 10^{-4}$) and libero-10, and the gain follows need: on spatial, whose checkpoint already survives 90% of shifts, it is neutral. On $\pi_{0.5}$ one rollout per task does not resolve the camera channel ($p = 0.18$), so we ran the same 30 camera tasks three times: over 90 paired episodes the wrapper repairs 21 and breaks 8 ($0.22 \to 0.37$, $p = 0.024$).

Baselines (Table 3). Unmeasured corrections do little, and the measured pose without depth is not enough. Test-time augmentation is a real baseline that breaks nothing, but over 150 camera tasks on three suites the wrapper beats it on 46 tasks and loses on 13 ($p < 10^{-4}$), and augmentation alone does not beat the raw policy on libero_object or libero_goal.

4.3 The cost of measuring, and a test that removes it

Self-measurement moves the arm, and the held-out $\pi_{0.5}$ is sensitive to where its episode starts. On the ten-task sweep the measuring wrapper repairs 44 of 82 $\pi_{0.5}$ failures but breaks 45 of 158 successes, 40 of them on units where no camera correction was applied: the

Table 2: Paired results, per-suite OFT checkpoints and the held-out $\pi_{0.5}$ (rec.: repaired of raw failures; broken: of raw successes). The 240 camera tasks are the 60 libero_10 tasks and 60 on each other suite.

policy	suite	joint offset			camera		
		raw → wrapped	rec.	broken	raw → wrapped	rec.	broken
OFT	spatial	0.25 → 0.95	15/15	1/5	0.90 → 0.88	3/6	4/54
OFT	object	0.20 → 1.00	16/16	0/4	0.42 → 0.75	26/35	6/25
OFT	goal	0.20 → 1.00	16/16	0/4	0.67 → 0.78	12/20	5/40
OFT	libero-10	0.33 → 0.90			0.35 → 0.63	20/39	3/21
$\pi_{0.5}$	libero-10	0.37 → 0.90	16/19	0/11	0.20 → 0.37	7/24	2/6

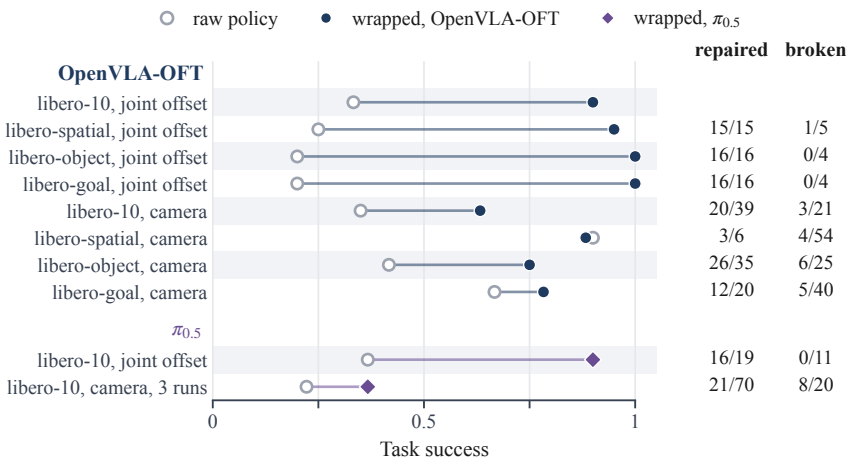


Figure 4: Task success of the raw policy (hollow) and the same policy wrapped (filled) on the same tasks and seeds; right: raw failures repaired and raw successes broken. The $\pi_{0.5}$ camera row pools three replicated runs (Sec. 4.2). Baselines are in Table 3.

sweep hands the arm back 10 to 25 mm from where the episode began and still moving at 0.19 rad/s. The raw policy re-run with the same seeds changes outcome on 0 of 20 tasks, so this is the arm, not sampling. Prop. 3 removes it†: the robot looks once at its own silhouette without moving and measures only if the silhouette misses the training camera's prediction. The test separates the cases on every logged unit (sweep: residual 0.00 on all 60 unshifted units and above 1.9 on all 70 shifted ones; LIBERO-Plus: below 0.02 on all 140 unshifted tasks and above 3.0 on all 240 camera-shifted ones). Run over the whole sweep, the motion-free arm breaks 7 of 138 raw successes where the measuring wrapper breaks 40, keeps 21 of its 25 repairs, and beats it on 41 units against 12 ($p = 8 \times 10^{-5}$). Against the raw policy it is a net gain, 21 repaired against 7 broken ($p = 0.013$), where the measuring wrapper was a wash (25 against 40). When the test fires, the arm measures and then settles where the episode began; on OFT camera tasks the settled arm beats the plain one on 7 tasks and loses on 2 over 90.

4.4 Where the deficit lives

When a corrected policy still fails, which part failed? We answer by intervention†: on the same task and seed, four nested arms each remove one more cause (raw; measured pose; the simulator's true pose; a perfect render from the training camera), and a raw failure is attributed to the first arm that repairs it.

No raw camera failure on these tasks survives a perfect undo, and the ones only a perfect render repairs are the ones the robot had already scored as lost view (lowest coverage). Extending the upper arms to every double failure on all four suites, 100 raw camera failures split into 61 repaired by measurement, 17 only by the true pose, 21 only by a perfect render,

Table 3: Baselines on 30 libero_10 camera tasks, every arm on every task (OFT).

arm	success	repaired	broken	wrapper vs. arm (wins / losses, p)
raw policy	0.27			
generic corrections (white balance, CLAHE, denoise)	0.37	3/22	0/8	10 / 2, 0.039
homography with our pose, no depth	0.40	9/22	5/8	9 / 2, 0.065
test-time augmentation	0.47	6/22	0/8	8 / 3, 0.23
wrapper	0.63	13/22	2/8	

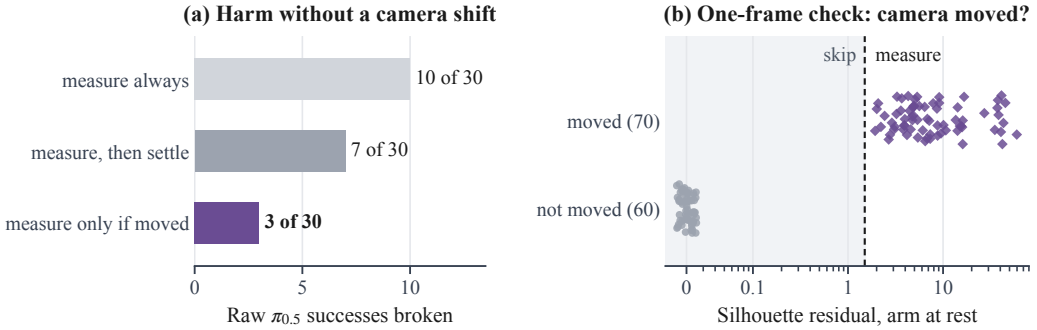


Figure 5: (a) Raw successes of $\pi_{0.5}$ broken by the wrapper on 40 tasks where it corrects nothing: measuring always, measuring then settling, and measuring only when the silhouette says the camera moved. (b) That signal: the silhouette residual at the training camera from one rest pose, unshifted against camera-shifted units.

and one by nothing: the policy is almost never what fails. Calibration is bimodal (the units only the true pose repairs are gross failures, median 39 cm; the units only a render repairs were calibrated to 2.2 cm), and a rollout-free probe puts a scale on it: both families leave their training behaviour at about 1 to 2 cm and 1 degree of pose error (Fig. 6), and self-calibration sits inside that (0.07 to 0.54 cm). Among raw camera failures, self-measured coverage predicts repair at AUROC 0.74 [0.64, 0.84] ($n = 98$) where the size of the shift does not (0.52 translation, 0.66 rotation). On LIBERO-Plus level-5 tasks, which no reference VLA solves, the servo repairs 32 of 33 robot-initial-state failures, and within level-5 camera failures, where the benchmark's grade is constant, coverage still ranks what is fixable (0.75, $n = 47$): hard for every policy is not unrepairable.

4.5 What the policy's own actions say

Queried at seven end-effector positions the robot knows from its encoders, a policy's commanded motions should meet at a sub-goal if it understands the frame. They do not, on either family (AUROC 0.32 to 0.56), and Prop. 4 says why: on OFT the symmetric part of the fitted action Jacobian has positive trace on 97% of units and the triangulated point lies behind the gripper on 94%. At the start of an episode a behaviour-cloned policy is a source, reading a displaced arm as further along a demonstration fan, contrary to the contracting fields that stable dynamical-system policies assume (Beik-Mohammadi et al., 2024). What the same probes do carry is a fingerprint: the distance between the seven probe actions under a frame and under the same task in its training setup (Table 5).

As a predictor the fingerprint is strong where the policy is coherent in its training setup, and the probing is what carries it: on the two families with that property, seven probes reach 0.93 and 0.79 against 0.57 and 0.68 for the opening action alone, and on OFT the AUROC rises with the number of probes (0.80, 0.84, 0.93 for one, three and seven). As a per-frame selector it helps only on $\pi_{0.5}$, where it kept the raw frame on all five units a correction would have broken and chose better than always correcting (0.86 against 0.81). Base OpenVLA never matches its own training reference (none of its units comes within the frozen 0.10 of it), so the statistic has nothing to compare against; the registered test fails there (Appendix A). Across the registered decision rules, predicting what is repairable proves easier than deciding frame by frame when to repair: a do-no-harm veto and a coverage threshold also did no better than always correcting, and are reported with the fingerprint in Appendix A.

Table 4: Attribution ladder, OFT, the first 30 libero_10 camera tasks. Bottom: the 17 raw failures by the first arm that repairs them, with the coverage the robot measured before acting.

	raw	measured pose	true pose	perfect render
success (of 30)	13	19	21	27
first arm that repairs a raw failure	–	7	2	8
reading		repaired	not identified	not invertible
self-measured coverage, median		0.58	0.96	0.44

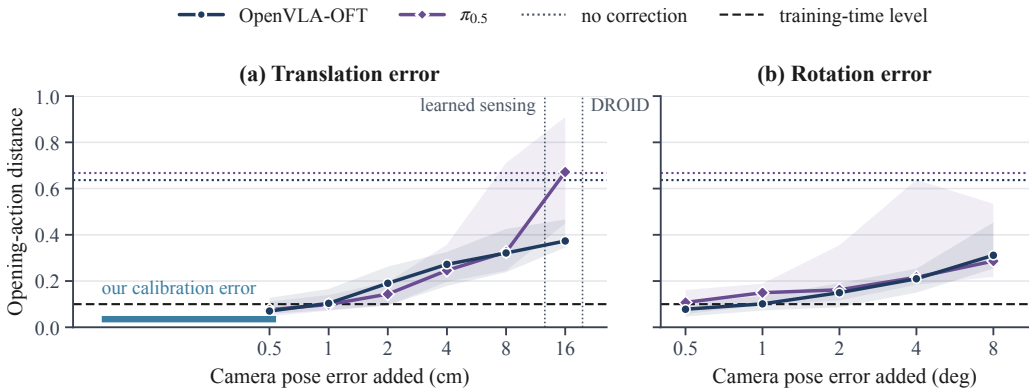


Figure 6: How exact the calibration must be: the policy's opening-action distance to its true-pose action as a known pose error is added in translation (a) or rotation (b), two frozen families, no rollouts; median and interquartile range over tasks. Dashed: the frozen fingerprint threshold; dotted: no correction. Marked on the same axis: the wrapper's own calibration (bar, 0.07 to 0.54 cm), learned sensing (12.6 cm) and real DROID images (19.6 cm).

5 Limitations

Sensing. The main results use simulator segmentation and depth, and we measured what replacing them requires: the policies tolerate about a centimetre and a degree of calibration error. Learned robot masks and depth reach 12.6 cm on the same 30 camera tasks and real DROID frames 19.6 cm (Appendix B), both outside that tolerance, and accordingly learned sensing does not yet improve on the raw policy (12 of 30 against 13; 19 with simulator sensing). The contribution on this axis is the target a sensing pipeline must meet, stated as a number. Scope. All policy evaluation is in simulation on one benchmark family with three policies; DROID tests the self-measurement and the self-check on real images ($\rho = -0.40$ with true error, $p = 0.003$), not a policy. Design. The invert terms are designed per channel, fixed before any held-out data, and coverage is geometric. A nuisance the body does not model degrades gracefully: an unmodelled 20% zoom is absorbed as a camera backing along its own axis and half of the correction's benefit survives, and focal length is identifiable from three poses (not aliased with the pose), so it can be added as a channel (Appendix C). A calibration takes about a minute, which suits pause-and-measure deployment rather than per-frame use.

6 Conclusion

Before asking a policy to be robust to a shift, ask whether the shift is the policy's problem. A robot can answer that about itself: what it can measure with its own body it can often undo, how much of the undoing will work is predictable without running the policy, and whether it needs to measure at all it can tell by looking.

Table 5: Action-field fingerprint; constants frozen on OFT before any held-out probe. Row 3: success of always raw, always wrapped, the frame the fingerprint picks, and the per-unit oracle.

	OFT (dev.)	$\pi_{0.5}$ (held out)	base OpenVLA (held out)
AUROC, corrected-frame distance predicts wrapped success	0.928	0.787	0.643
the same from the opening action alone (no probes)	0.571	0.679	0.693
raw / wrapped / fingerprint / oracle	0.70 / 0.84 / 0.80 / 0.90	0.70 / 0.81 / 0.86 / 0.90	0.13 / 0.46 / 0.32 / 0.48
units a correction broke, kept raw	0 of 1	5 of 5	0 of 1

AI use statement

We used generative AI tools (large language model coding and writing assistants) to write and debug experiment code, to recompute and cross-check the reported statistics from the logged result rows, to search for related work, and to draft and edit text. A vision-language model is also part of one experiment, where it chooses robot mask regions on DROID images (Appendix B); there it is a method component. Every reported number was recomputed from the released result rows with the analysis scripts, and every reference was checked against its arXiv, proceedings or publisher record. We have reviewed all AI-assisted work and take responsibility for the final content of this paper, including text, claims and artifacts produced with the aid of generative AI.

Reproducibility statement

Every constant of the score, the fingerprint and the registered decision rules was written to a dated log before the data that tests it existed, and each registered test is reported against that entry, whatever its outcome (Appendix A). Proofs are in Appendix D. Code, the run notebooks and every result row will be released.

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A Registered tests and their outcomes

Each test below was written to a dated log, with its constants and pass criteria, before any data that could test it existed. Development-family (OFT) outcomes are descriptive; held-out outcomes are the test.

B Real images: DROID

Offline calibration of DROID from robot masks is not new (Xie et al., 2025; Irshad et al., 2025). What we add is why the obvious pipeline fails and what repairs it: completing a mask with frame-to-frame motion re-attaches moving background unless motion is trusted only inside regions already called robot (precision 0.91 falls to 0.75 otherwise), and keeping the largest connected component drops parts of an arm cut by the frame edge (recall 0.68 to 0.53 on one camera). The ground-truth-free self-check correlates with the true error at $\rho = -0.40$ ($p = 0.003$, $n = 53$). A dedicated robot segmenter (Mei et al., 2025) would be a natural third mask source; the comparison here holds the search fixed and varies only the mask.

test (registered)	rule	outcome
Recoverability score as a predictor (frozen 2026-09-17)	rank raw failures by recover	passes on both held-out families: 0.809 ($\pi_{0.5}$), 0.675 (base OpenVLA); category label 0.515, 0.472
Do-no-harm gate (2026-09-18)	veto a correction if the action changes more under benign jitters of the corrected frame than of the raw one	fails: over 386 paired gated / ungated units on the two held-out families it vetoed 49, saved 1 unit and broke 16 ($p = 3 \times 10^{-4}$); where it does not veto it reproduces the ungated arm exactly
Score-gated correction (2026-09-19)	correct the camera only if coverage ≥ 0.5	killed by its own criterion on rows unseen when frozen: prevented 2 breaks, gave up 3 repairs
Ray-concurrency probe (2026-09-19)	triangulate a sub-goal from 7 probe actions	fails on both families (AUROC 0.32 to 0.56); explained by Prop. 4
Action-field fingerprint, H-F1 to H-F5 (amendment, 2026-09-19, before any held-out probe)	H-F1 corrected AUROC ≥ 0.80; H-F2 raw AUROC ≥ 0.70; H-F3 keep raw on $\geq$ half the units a correction breaks; H-F4 keep the correction on $\geq 80\%$ of units it repairs; H-F5 selected success $\geq$ best fixed arm minus one unit	$\pi_{0.5}$: H-F3 and H-F5 pass (5 of 5; 0.86 against 0.81), H-F1 0.787, H-F2 0.684, H-F4 7 of 9 narrowly fail. Base OpenVLA: H-F1 0.643, H-F2 0.659, H-F4 3 of 13, H-F5 0.32 against 0.46 fail; H-F3 undetermined (one unit)
DROID self-calibration stop rule (written before the final calibration run)	stop if the median error on reliable references exceeds 30 cm	fired (32.1 cm); the later mask change reaching 19.6 cm is post hoc (Appendix B)

Table 6: Self-calibration on 53 real DROID cameras; identical search, only the mask differs. The mask was changed after the registered stop rule had fired on the first attempt (post hoc).

	median error	under 30 cm	under 20 cm	paired Wilcoxon
zero-shot masks	25.4 cm / 25.1°	28	19	
VLM-chosen masks	19.6 cm / 19.3°	40	28	$p = 0.015$

C Additional results and figures

Rollout noise on $\pi_{0.5}$. Across the three replicated runs of Sec. 4.2, the outcome of a task changes between runs on 10% of camera tasks under the raw policy and 33% under the measuring wrapper, the arm-motion mechanism of Sec. 4.3 seen as variance. On the 10 light tasks run alongside, the measuring wrapper breaks 13 of 26 raw successes and repairs 3 ($p = 0.021$); the motion-free arm of Sec. 4.3 breaks 1 of 54 light successes on the ten-task sweep.

A nuisance the body does not model. Zooming the true camera while the wrapper assumes the training focal length (8 camera tasks, policy-free) is absorbed as a camera that backed away along its own optical axis: at 5, 10 and 20% zoom the pose error is 2.7, 5.9 and 10.8 cm, almost all of it along that axis. The corrected frame degrades gracefully (image error 14.5, 16.3 and 20.1 against about 30 uncorrected and 10.2 with the true focal length), and the self-check barely notices (F1 0.93 to 0.83): it verifies the pose it was asked about, not the model behind it. Focal length is not aliased with the 6-DoF pose (Jacobian residual 0.78 against a median of 0.51 among the pose columns), so three poses at different depths identify it; lens shift in x is aliased with a 0.44 degree pan, which undoes it.

D Proofs

Proposition 1. Linearise $z(q_k, \eta)$ about the true η: $z_k = z(q_k, \eta) + J_k\delta + \epsilon_k$ with $\epsilon_k \sim (0, \sigma^2 I)$ independent. Stacking gives a linear Gaussian model with Fisher information $\sigma^{-2} J^\top J$. The Cramér–Rao bound gives $\mathrm{Cov}(\hat{\eta}) \succeq \sigma^2 (J^\top J)^{-1}$, so $\mathbb{E}\|\hat{\eta} - \eta\|^2 \geq \sigma^2 \mathrm{tr}((J^\top J)^{-1}) \geq \sigma^2 \lambda_{\max}((J^\top J)^{-1}) = \sigma^2 / \sigma_{\min}(J)^2$. Appending rows to J appends a positive semidefinite term to $J^\top J$, and by Weyl's inequality its smallest eigenvalue cannot decrease. The bound is first order; the calibration we run also uses depth, which adds rows to J and can only

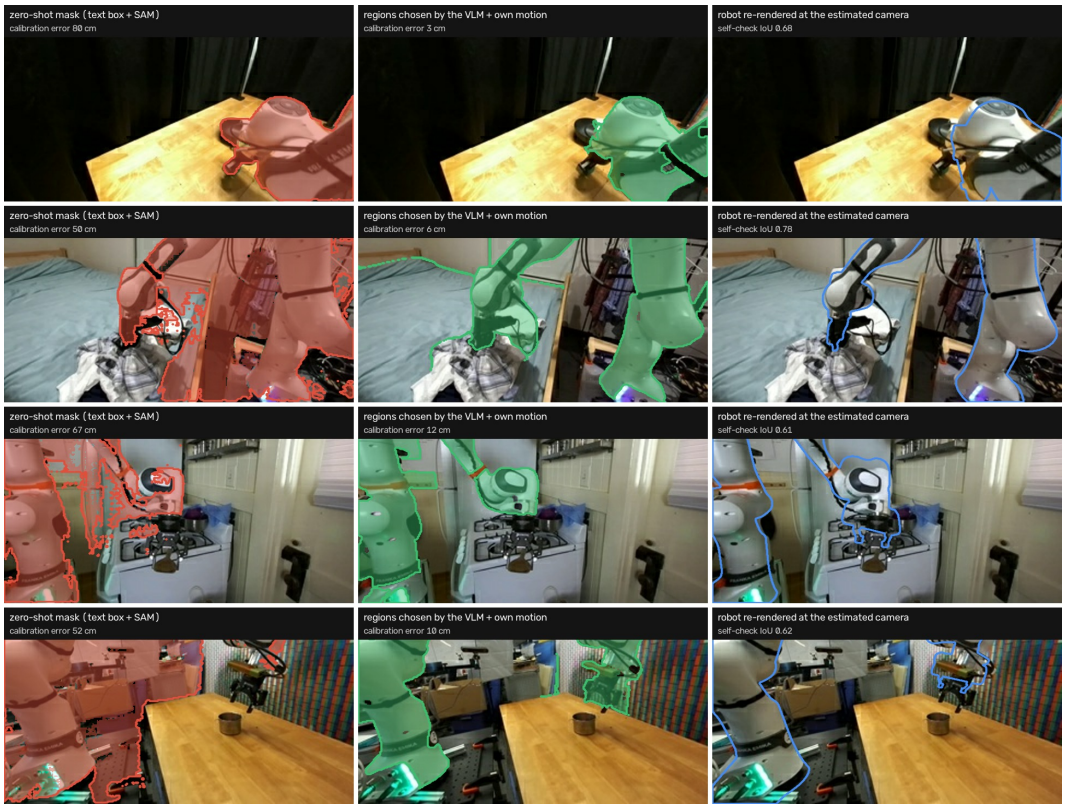


Figure 7: Real DROID frames. Left: a zero-shot text-box-plus-SAM mask attaches background to the robot, and a silhouette calibration follows it. Middle: SAM regions chosen by a vision-language model, with part-robot regions split by the robot's own motion. Right: the robot re-rendered at the estimated camera: the agreement is a check the robot can run with no ground truth.

Table 7: The first score test, three base tasks per family (the setting in which the score was first frozen).

	units	repaired	AUROC, score	AUROC, label
OpenVLA-OFT (development)	21	14	0.908 [0.68, 1.00]	0.500
$\pi_{0.5}$ (held out)	21	14	0.837 [0.64, 1.00]	0.464
pooled	42	28	0.881 [0.76, 0.98]	0.482

tighten it. (Measured with the full excursion of the 7-pose sweep; halving the excursion gives 1.01 to 3.23 and 0.54 to 0.18 cm.)

Proposition 2. Split $\Omega = V \cup (\Omega \setminus V)$. A pixel $u \notin V$ has no pre-image in the deployed frame o_d, so $o_{\text{train}}(u)$ and o_d are conditionally independent given the scene prior, and any $\hat{o}(u) = f(o_d)$ has the Bayes risk of the best predictor from the prior alone, $e_{\text{prior}}(u)$. Summing over $\Omega \setminus V$ and adding the resampling error on V gives the bound.

Proposition 3. With $\eta = \eta_0$ and an exact body model, the rendered and observed silhouettes are the same function of q, so $r(q) = d(\text{noise})$. If $r(q)$ exceeds what noise alone produces, the premise fails. The test is one-sided: a zero residual at one pose is consistent with a small set of poses that project the arm identically at that q, which is why the method measures with several poses once the test fires.

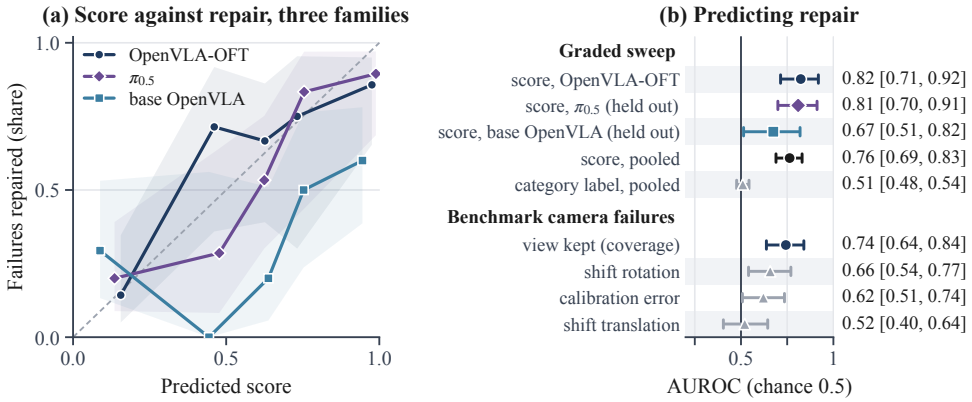


Figure 8: (a) Observed repair rate against the policy-free score on the graded sweep, binned with Wilson bands, for the development and both held-out families. (b) AUROC with 95% bootstrap intervals: the score against the category label on the sweep, and, among benchmark camera failures, coverage against the size of the shift and the calibration error.

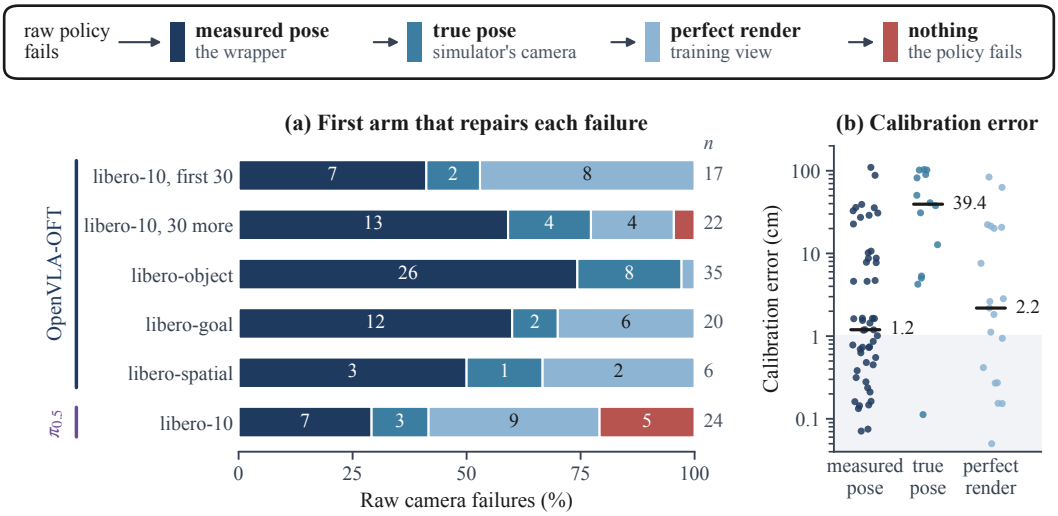


Figure 9: Top: the four nested arms, run on the same task and seed. (a) Every raw camera failure attributed to the first arm that repairs it; pooled over OFT, 1 of 100 belongs to the policy. (b) The calibration error per bucket (bars: medians; shaded: below the policy's measured tolerance of about 1 cm): only-true-pose units are gross calibration failures, only-render units were calibrated to about 2 cm and lost the view.

Proposition 4. Under the triangulation model $a(p) = \lambda(g - p)$ the field's Jacobian is $-\lambda I$ and g its attracting fixed point. For the observed field $a(p) = 0$ at $g^\star = p_0 - A^{-1}a_0$, and the flow $\dot{p} = a(p)$ linearised there has Jacobian A; if $\frac{1}{2}(A + A^\top) \succ 0$ then $\frac{d}{dt}\|p - g^\star\|^2 = 2(p - g^\star)^\top A(p - g^\star) > 0$, so $g^\star$ repels. The least-squares triangulation returns $g^\star$ regardless, and its residual measures the asymmetry of A, not the distance between the observation and the training distribution.

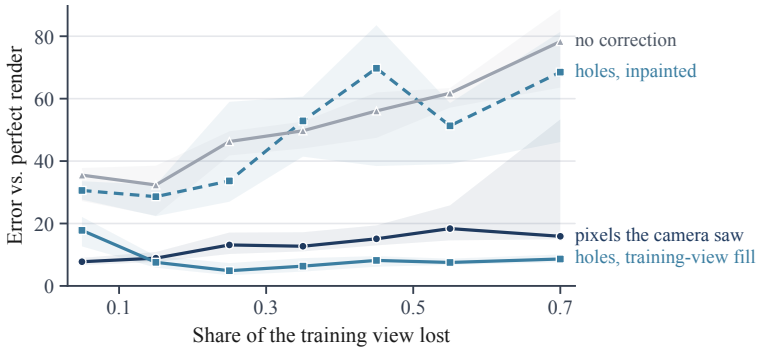


Figure 10: Error against a perfect training-view render on the 264 placements, split into seen pixels and holes, as a function of the view lost (Prop. 2); median and interquartile range. With a training-view fill the holes are cheap; the error that grows with lost view sits on the seen pixels.



Figure 11: The motion-free test of Prop. 3: the silhouette the training camera predicts for the current joint configuration (blue) over the deployed frame. When the camera has not moved it lands on the arm and the sweep is skipped; when it has, the miss is visible before the arm moves.

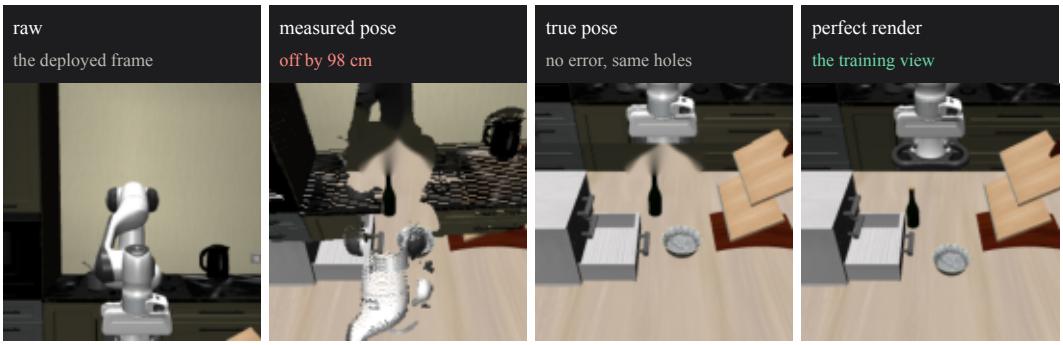


Figure 12: The ladder on one task (camera turned away 35 degrees): the measured pose fails, the true pose still leaves the view the camera lost, only the training camera's own view is complete.

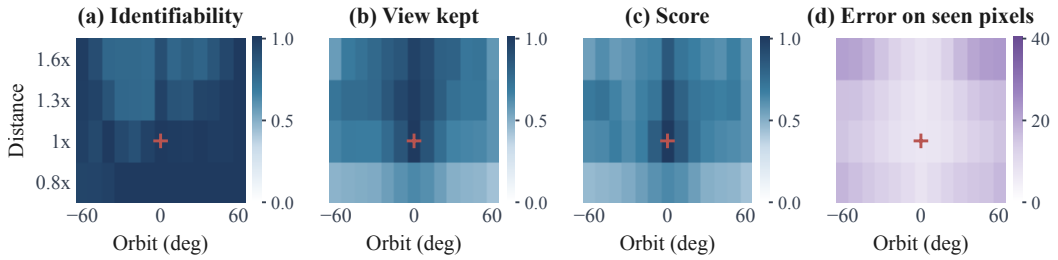


Figure 13: Policy-free maps over camera placements around the training camera (cross), means over three base tasks: identifiability, view kept, the score, and the seen-pixel error after a true-pose re-rendering.

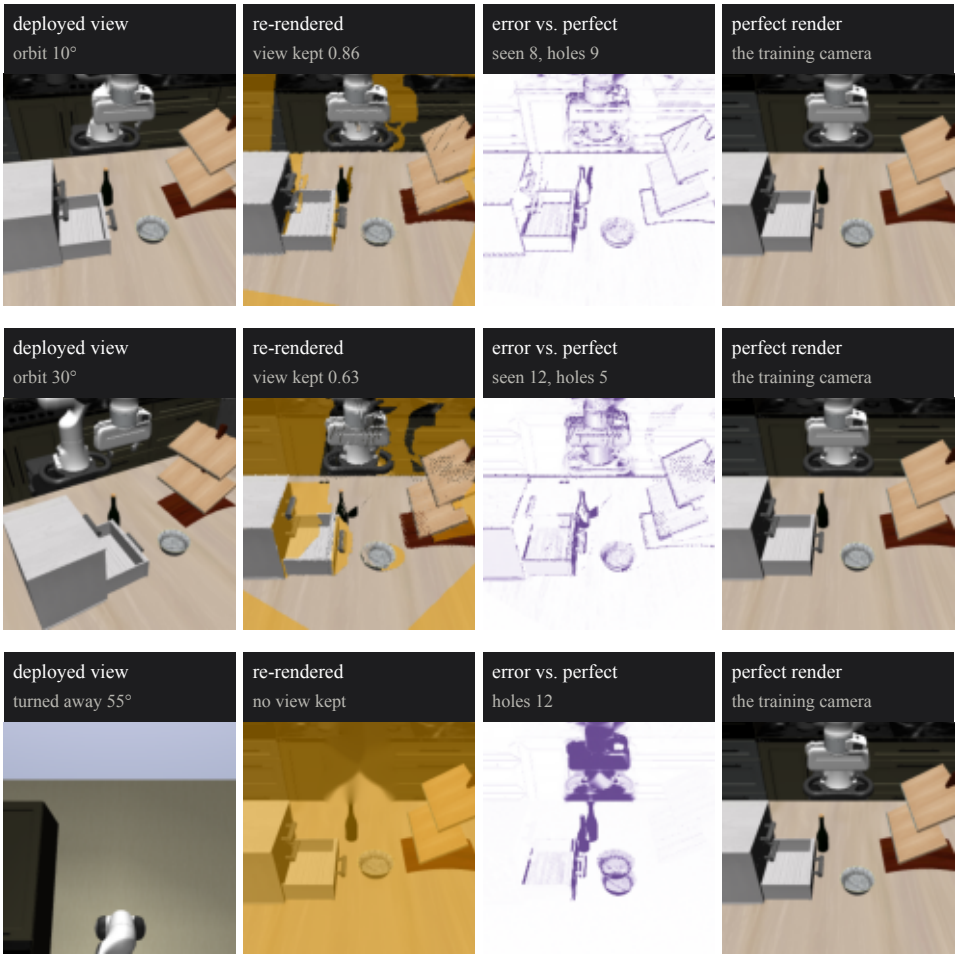


Figure 14: Where the corrected frame is wrong, in pictures: holes (yellow) are filled cheaply; the error sits on seen surfaces and grows with the orbit.

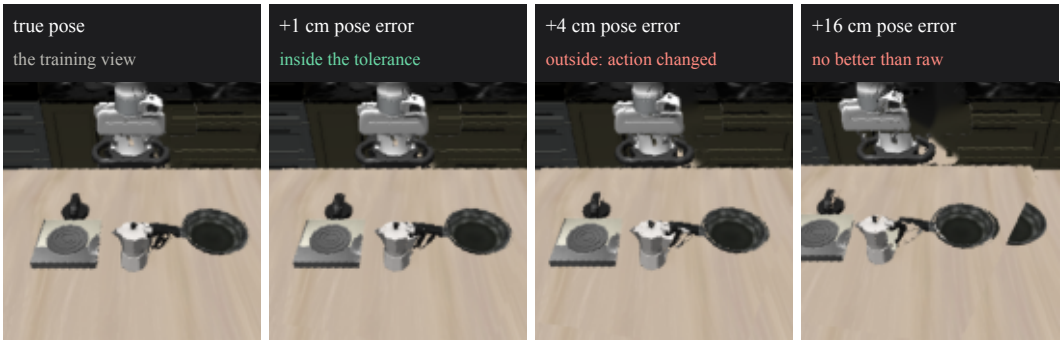


Figure 15: The frame handed to the policy when the true camera pose is perturbed by 1, 4 and 16 cm.